Sensor faults and actuator commands in a low-cost closed hydroponic system: a retrospective audit

MD. Sameer Iqbal Chowdhury[a], Mikdam-Al-Maad Ronoue[a], Md Asaduzzaman[b], Lafifa Jamal[a,*]

[a]*Department of Robotics and Mechatronics Engineering, University of Dhaka, Dhaka-1000, Bangladesh*
[b]*Department of Agriculture and Food Technology, Kyoto University of Advanced Sciences, Japan*

Abstract

We audit how sensor faults can propagate into actuator commands in a low-cost Internet of Things hydroponic controller. A 30-day closed nutrient film technique lettuce deployment produced 121,244 logged samples, including impossible pH values, prolonged full-scale readings, and gaps covering 23% of the period. Replay combines the reported setpoints and executable actuator directions with a common 10-min cooldown. Under naive control, flagged readings supplied 19.1% of pH dosing commands and 10.2% of CO_2 injection commands. A physical-range check with 32 s persistence reduced these fractions to zero and 2.0%. Flags distinguish out-of-range values, bound readings, and heuristic spikes; they are not independent fault diagnoses. Only 3% of valid air-temperature readings and 10% of valid water-temperature readings were in band, indicating poor regulation with an unresolved cause. LiveGate, a forecast-consistency and sensor-liveness gate, did not outperform simple baselines overall under a frozen retrospective protocol. A post-hoc comparison with run-length and bound-rejection gates exposed substantial availability costs. Learned liveness reduced commands using injected stuck readings by 79%, but useful-command coverage in those trials fell from 98% to 27%. Commands were reconstructed, not observed; actual dosing, prevented damage, and crop benefits remain unmeasured.

Keywords: hydroponics, sensor faults, actuation, retrospective audit

*Corresponding author.

1. Introduction

Closed hydroponic systems recirculate a nutrient solution whose pH, electrical conductivity (EC), and temperature must be held within narrow ranges, while the surrounding air temperature, humidity, and carbon dioxide (CO_2) concentration shape growth in enclosed production [1, 2]. Automated controllers built from low-cost sensors, microcontrollers, and single-board computers now regulate these quantities with dosing pumps, valves, chillers, and coolers [3, 4]. In such controllers a sensor reading is not only a data point: it can trigger a physical action. One spurious high pH reading can authorize acid dosing into the whole recirculating volume, and the correction may require dilution, counter-dosing, and another mixing interval.

System studies of IoT hydroponics describe architectures, dashboards, and crop outcomes, while sensor-fault studies characterize faults and their detection [5, 6, 7, 8]. Hydroponic fault detection predates the current IoT architectures [9], and SensorTalk connects agricultural fault diagnosis to relationships between sensors and actuators [10]. The narrower question here is how flagged readings change executable command counts under a specified controller, and how much useful authorization is lost when readings are rejected. This requires accounting for actuator direction, persistence, cooldown, and the age of held readings together.

This paper examines this narrower question with data from a deployed closed NFT lettuce system. Its contributions are:

1. a descriptive inventory of invalid, spiking, repeated, and missing readings in a 30-day hydroponic deployment;
2. an executable-command replay that attributes authorizations to their source readings and compares six gating rules;
3. a frozen retrospective comparison of LiveGate with 14 alternatives, reporting its negative overall result, followed by a separate comparison with simple run-length and physical-bound gates;
4. engineering implications for gating, instrumentation, and temperature regulation, with explicit limits on physical and agronomic inference.

2. Related work

2.1. IoT control of hydroponic systems

Automated hydroponic control typically measures pH and EC in the nutrient tank and corrects them with acid, base, concentrate, or water dosing,

often with a mixing delay between doses [3]. IoT variants add networked logging and remote monitoring [4]. Reliability in these studies is usually discussed in terms of communication and uptime, and evaluation focuses on whether the controlled variables stay near their setpoints and on crop performance. How often the controller acts on readings that were wrong is rarely reported.

2.2. Sensor faults and their detection

Field sensor deployments exhibit outliers, spikes, stuck-at values, offsets, drift, and missing data [5, 6], and data quality in IoT pipelines has been reviewed systematically [7]. Reviews of agricultural IoT report similar fault classes and a growing set of model-based and learning-based detectors [8]. Classical detectors include robust outlier filters [11], sequential change detectors such as CUSUM [12, 13], exponentially weighted moving averages [14], state estimation with the Kalman filter [15], and density-based novelty detection such as the local outlier factor (LOF) [16]; model-based fault diagnosis is surveyed in [17]. Range and local-change tests can accept a plausible constant trace without independent evidence of process change. Persistence-based confirmation can then reinforce an erroneous authorization.

2.3. Conformal calibration

Split conformal prediction converts a predictor into a set or threshold with a finite-sample coverage guarantee under exchangeability [18, 19]. Here its quantile rule provides a nominal calibration parameter. We make no formal coverage guarantee for these dependent sensor time series; the nominal parameter is not a guaranteed clean-reading rejection rate, and performance is assessed empirically.

2.4. Diagnosis and consequences in controlled environments

Neural-network detection of mechanical, sensor and biological faults has been studied in deep-trough hydroponics, including a proposed real-time supervisor [9, 20]. Prior approaches already use control information: the neural-network study includes actuator control signals among its inputs, while SensorTalk temporarily feeds standard-sensor readings to actuators during calibration [9, 10]. More recent work addresses EC and pH fault diagnosis in hydroponic systems [21]. In controlled-environment agriculture, parity residuals and observers use identified process relationships to isolate faults,

subject to model and fault assumptions [22]. These diagnosis questions require different information and endpoints from retrospective command attribution; classification accuracy cannot be ranked against command coverage across different datasets. Sensor-error consequences also have a precedent: reference-measured greenhouse errors were used to simulate production and energy effects [23]. Our contribution is therefore a deployment-specific authorization audit with explicit source provenance and executable directions, rather than a new general principle of agricultural fault diagnosis or evidence of measured physical consequences.

3. System and deployment

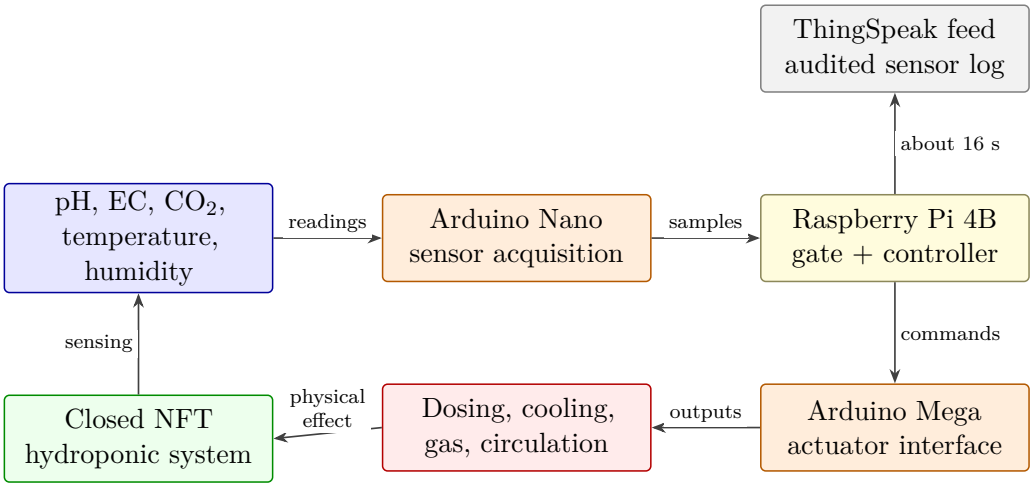
3.1. Platform

The platform is a closed NFT system for leaf lettuce consisting of a Growth Module and a Nutrient Solution Container Module (Fig. 1, Table 1). The Growth Module measures air temperature and humidity with a DHT22 sensor and CO_2 with an MH-Z19C non-dispersive infrared sensor whose range ends at 5000 ppm. CO_2 is raised by a normally closed solenoid valve on a CO_2 cylinder, air temperature is lowered by a thermoelectric cooler, and LED grow lights provide the photoperiod. The Nutrient Solution Container Module measures pH with an Atlas Scientific analog pH kit, EC with an Atlas Scientific Mini Conductivity K 1.0 kit, water temperature, and water level with a JSN-SR04T ultrasonic sensor. pH is corrected by acid or base dosing, EC by concentrate or distilled-water dosing, and water temperature by a refrigerative chiller. A pump circulates the solution continuously and an aerator runs throughout.

Sensors are read by an Arduino Nano, which applies median filtering; actuators are driven by an Arduino Mega; a Raspberry Pi 4 Model B runs the controller and uploads readings to a ThingSpeak channel at about 16 s intervals. Actuator commands and relay states were not logged.

3.2. Replay control and the legacy-gate proxy

The system description specifies a correction when a reading leaves its setpoint band and a 10-min mixing lockout after pH or EC dosing. The replay uses that interval as a common cooldown for all five channels; its extension to CO_2 injection and cooling is a comparison assumption, not verified deployment timing. In the first growing season the controller acted on logged readings without further gating.



Command and relay states were not logged.



Figure 1: Sensing-to-actuation path (top) and the deployed NFT system (bottom). Sensor readings were logged; command and relay states were not recorded.

Table 1: Controlled quantities, deployed setpoint bands, and the directions the installed actuators can execute. Humidity and water level were monitored but not actuated.

Quantity	Setpoint band	Actuator	Executable direction
pH	5.8–6.5	Acid / base dosing	Both
EC	1000–1500 $\mu S\,cm^{-1}$	Water / concentrate dosing	Both
CO_2	1900–2100 ppm	Solenoid valve on cylinder	Raise only
Water temperature	18–22 °C	Refrigerative chiller	Lower only
Air temperature	20–26 °C	Thermoelectric cooler	Lower only

For the second season, the description gives a sensing rule that accepts a reading when it differs from the last trusted value by at most $(1 + \beta)\xi$, with a reset after 50 consecutive rejections. It also describes $\alpha\xi$ actuation bands around the setpoint limits and a repeatability count ψ, but the band notation does not uniquely specify executable firmware behavior. Our legacy-gate proxy retains the sensing/reset rule, omits the α bands, and uses 16ψ seconds of out-of-band persistence of an age-eligible held estimate in the common controller. Both ξ and ψ are re-estimated from development data because their original first-season inputs are unavailable. This arm is an approximate comparison, not demonstrated firmware reconstruction. The published multipliers β were 0.0002 (EC), 0.0005 (pH), 0.0022 (air temperature), 0.0009 (water temperature), and 0.0010 (CO_2).

3.3. Deployment

The analysed deployment ran from 27 February to 28 March 2024 in Dhaka, Bangladesh, with ambient temperatures of about 25–36 °C, which is off-season for lettuce. Seven seedlings per treatment were grown in the NFT system, in pots with soil and compost, and in a manually managed deep water culture (DWC) system.

4. Methods

4.1. Data

The log contains 121,244 rows with a median interval of 16 s. The analysis uses the five actuated quantities, plus humidity for the fault inventory. Water level is excluded because its raw values were never calibrated to a physical level. Timestamps are split chronologically into development (first 40%), validation (next 30%), and evaluation (last 30%) blocks.

4.2. Fault inventory

For each channel we count readings outside the physical range (0–14 for pH, 0–5000 $\mu S\,cm^{-1}$ for EC, 250–5000 ppm for CO_2, −10–60 °C for air and 0–50 °C for water temperature, 0–100% for humidity); isolated spikes, defined as in-range readings whose deviation from a centred 9-sample median exceeds eight robust standard deviations; flat-line runs of identical readings lasting at least 1 h without a logging gap; the fraction of readings at the channel's in-range maximum; and logging gaps longer than 120 s. These are descriptive definitions, not ground-truth labels.

4.3. Fault-to-actuation audit

The audit replays the reported setpoint bands and a specified common 10-min cooldown on the unmodified log under six gating arms: (i) a naive threshold on every logged reading, as in the first season; (ii) a physical-range check; (iii) a range check with 32 s persistence before acting; (iv) the approximate legacy-gate proxy described above; (v) arm (iii) plus the LiveGate liveness test (Section 4.6); and (vi) arm (iii) plus rejection of readings exactly at a declared physical bound. The last accepted value may be held for at most 64 s; a logging gap longer than 120 s invalidates it. Directions unavailable to the installed actuator are suppressed before persistence and cooldown are evaluated. An unavailable direction therefore cannot delay a later executable command.

Each command is attributed to the reading supplying its held estimate. We report three disjoint flag categories: out-of-range values, readings exactly at a declared bound, and in-range isolated spikes not already at a bound. A bound reading may represent a true excursion beyond the sensor's range, and a spike may be a legitimate transient; neither is an independently confirmed sensor fault. The replay uses a fixed historical background: its commands do

not change subsequent readings. It therefore measures authorization differences, not actual dosing, prevented damage, or crop effects. Band occupancy is the fraction of logged readings within range and away from declared bounds that fall inside the setpoint band; it is not a duration-weighted estimate for missing periods.

4.4. Post-hoc timing and flag sensitivity

After reproducing the original audit, we declared a separate sensitivity grid before executing it. We replayed the complete log with persistence of 16, 32, 64 or 96 s, maximum held age of 32, 64 or 128 s, and cooldown of 300, 600 or 1200 s. Naive and range-only controls retained zero persistence. Spike attribution used cutoffs of six, eight or ten robust scales; these descriptive cutoffs did not alter the online gate. The three arms, five channels and three spike definitions produced 1620 reported cells, including repeated control cells. Every cell was retained. The original setting had to reproduce before the full grid ran. These deterministic settings are sensitivity cases from one deployment, not independent replicates. Authorization retention is the count relative to naive replay at the same age and cooldown, not coverage of independently verified necessary actions.

4.5. Controlled benchmark

Because the log has no actuator records or reference instruments, faults that cannot be identified from the log alone are studied with controlled injection into real backgrounds. Eight non-overlapping, gap-free 240-sample windows are drawn from each of the validation and evaluation blocks. In each window, one of 12 scenarios is applied to one channel after a 60-sample warm-up: clean; positive or negative spikes, 30-sample biases, and 30-sample drifts; a 30-sample stuck value; 30 samples pinned at the declared physical maximum (saturation); a 30-sample dropout; and positive or negative legitimate level shifts that also change the reference background. Magnitudes are 0.5 for pH, 100 $\mu S\,cm^{-1}$ for EC, 2 °C for air and 1 °C for water temperature, and 500 ppm for CO_2.

Every method produces an accepted/rejected decision for each reading. A common downstream controller holds the last accepted value for up to a maximum age (64 or 320 s), requires out-of-band persistence (0, 32, or 96 s), and applies the 10-min cooldown. Reference opportunities are defined from the uncorrupted background with fixed 32 s persistence and the same cooldown; an authorization matches an opportunity if it occurs in the same

out-of-band episode and direction within 180 s. Coverage is the fraction of reference opportunities matched; undesirable authorizations are commands issued when the background is in band or in the opposite direction. The benchmark retains the frozen protocol's idealized two-direction controller for every channel; these counts are distinct from the executable-command audit.

Two secondary metrics distinguish authorizations occurring during injected-fault samples from authorizations whose last accepted source reading was fault-labelled. The former is the original protocol's temporal-coincidence metric; it does not establish use of a corrupted reading. The latter follows the held estimate's provenance, including commands after a fault ends. Neither metric alone establishes an undesirable direction relative to the background.

Each method's detector setting, persistence, and maximum age are selected on validation windows at three coverage targets (0.5, 0.8, 0.95): among settings meeting the target, the one with the fewest undesirable authorizations per trial is locked, and evaluation windows are then scored once. The 14 alternatives to full LiveGate comprise persistence only, Hampel, CUSUM, LOF, Kalman, EWMA, moving median, generalized likelihood ratio (GLR), three AASVR-R supervisor variants (measurement supervisor, hold-only, and without CUSUM), always-deny, and two LiveGate ablations. Kalman noise is scaled per channel from development data. Each split contains 480 channel–scenario trials derived from eight source windows; the evaluation windows span seven source days, not 480 independent deployments. Paired 95% percentile intervals resample those days jointly across sensors and scenarios with 1000 bootstrap replicates [24]. Seven clusters provide limited information about between-day variability. Component intervals are descriptive, without multiplicity adjustment; settings remain fixed during resampling.

4.6. LiveGate

LiveGate accepts a reading y_t for actuation only if it is consistent with a learned forecast and the sensor appears alive.

Consistency. Let ℓ_t be the mean of the last $K = 5$ accepted readings and τ_t the difference between the newest and oldest of them. The forecast is

$$\hat{y}_t = \ell_t + \mathbf{w}^\top \boldsymbol{\phi}_t + b, \qquad \boldsymbol{\phi}_t = \Big[\tau_t,\ \sin(2\pi h_t/24),\ \cos(2\pi h_t/24),\ \mathbf{z}_t\Big], \quad (1)$$

where h_t is the hour of day and $\mathbf{z}_t$ contains the other four actuated channels, winsorized at their development 1st and 99th percentiles and centred at their

medians. The weights are fitted by Huber regression [25] of $y_t - \ell_t$ on the first 75% of the development block, using only in-range, non-spike readings. The score is $s_t = |y_t - \hat{y}_t|/\sigma$, where σ is 1.4826 times the median absolute training residual. The threshold is the split-conformal quantile

$$q_\varepsilon = s_{(\lceil (n+1)(1-\varepsilon) \rceil)} \tag{2}$$

of the n clean scores in the last 25% of the development block.

Liveness. Let r_t be the length of the current run of identical readings. From the N runs of filtered readings in the first 75% of development, excluding readings at a declared physical bound, the smoothed survival function is $S(r) = (|\{\text{runs} \geq r\}| + 1)/(N + 1)$. The sensor is considered alive while $S(r_t) \geq \varepsilon$. The frozen implementation counts training runs after filtering and does not reset its liveness counter at logging gaps. The controlled comparison uses gap-free windows; whole-log liveness results remain subject to this gap-handling limitation.

Decision. A reading is accepted if it is within the physical range, the sensor is alive, and $s_t \leq q_\varepsilon$. A live but surprising reading starts a candidate level, and the trusted state re-anchors after

$$m(s_t) = \text{clip}\Big(1 + \lceil \log_2(s_t/q_\varepsilon) \rceil,\ 2,\ 8\Big) \tag{3}$$

mutually consistent readings, so larger surprises require more confirmation. The single parameter $\varepsilon \in \{0.002, 0.01, 0.05\}$ sets both the consistency and liveness thresholds and takes the place of the baselines' threshold multipliers.

Protocol. The method, the benchmark, and the decision rules were committed to the project repository before LiveGate was run on the measured log. LiveGate would be called best at a coverage target only if it had the lowest evaluation undesirable rate among methods feasible at that target, with coverage within 0.02 of the best feasible method; and best overall only if this held at every target where it was feasible. One design change, anchoring the forecast on a 5-sample level rather than the last value, was made beforehand to fix cascading rejections found in synthetic unit tests. The data had been analysed in earlier studies, so the evaluation is retrospective rather than on untouched data.

5. Results

5.1. Faults in the deployment log

Flagged readings were frequent and channel-specific (Table 2, Fig. 2). pH produced 464 physically impossible readings, up to 67,101, consistent with corrupted analog or serial transfers, although the fault location cannot be diagnosed from the log. CO_2 produced 819 out-of-range readings, up to 200,850 ppm, and 961 isolated spike flags; 2.3% of its samples were exactly at the 5000 ppm full scale, including one 8.5 h episode. Humidity read 99.9% for 99.97% of samples. Saturation, sustained near-condensation conditions, and a stuck or corrupted channel cannot be distinguished without independent measurements. EC and both temperature channels had fewer flags; this does not establish their calibration accuracy. All channels shared 194 logging gaps longer than 120 s, the longest lasting 9.0 h, together covering 23% of the deployment.

Table 2: Fault inventory of the 30-day deployment log (121,244 readings per channel). Spikes are in-range readings more than eight robust standard deviations from a centred 9-sample median.

Channel	Out of range	Maximum out-of-range	Spikes	Flat-lines ≥1 h	Longest (h)
pH	464	67,101	76	0	–
EC	0	–	14	0	–
CO_2	819	200,850	961	3	8.5
Air temperature	1	99.9	17	1	1.4
Water temperature	0	–	39	7	1.5
Humidity	0	–	10	79	35.2

The long CO_2 episode was preceded by a steady rise from about 2000 to 5000 ppm over 2 h (Fig. 3). The true concentration may have exceeded the sensor range, for example through a valve that failed open, but reference and valve-state measurements are absent. If the sensor was functioning and clipped at full scale, the reading gives a lower bound; if it was stuck, even that interpretation is unavailable. The replayed liveness test rejected the pinned readings within four samples.

5.2. Observed regulation

Among valid logged readings, 88% of pH, 84% of EC, and 47% of CO_2 readings were in band (Fig. 4). Only 3% of valid air-temperature readings

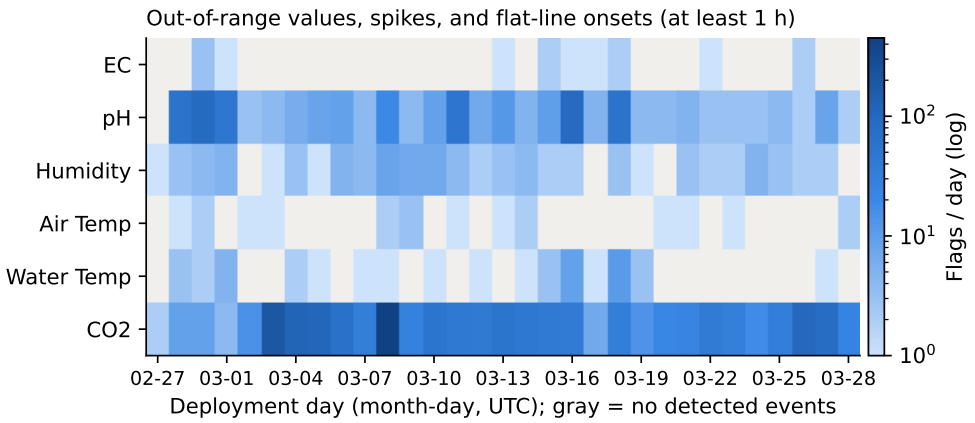


Figure 2: Daily descriptive flags: out-of-range readings, isolated spikes, and onsets of flat-lines lasting at least 1 h. These flags are not independent diagnoses of sensor faults.

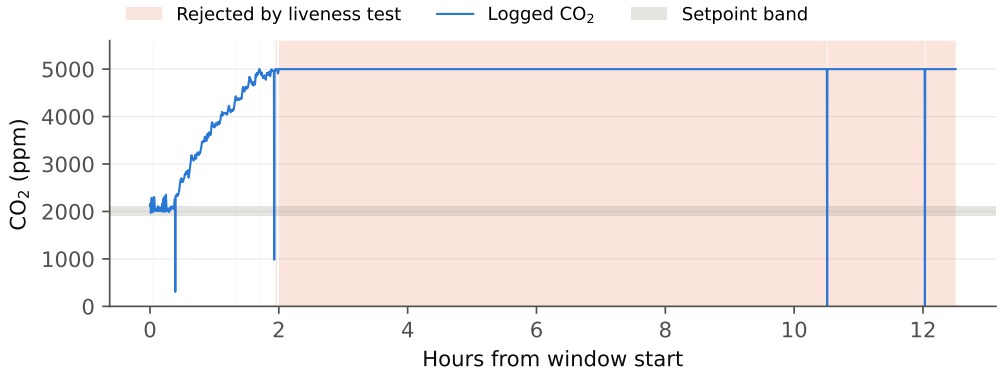


Figure 3: Longest CO_2 full-scale episode, with readings rejected by the replayed liveness test ($\varepsilon = 0.05$) shaded. Downward excursions are spike candidates; their physical cause is unverified.

were within 20–26 °C (median 28.6 °C), and 10% of valid water-temperature readings were within 18–22 °C (median 24.8 °C). These sample proportions do not describe the unobserved 23% of the deployment. The traces indicate poor temperature regulation, consistent with insufficient cooling capacity under hot-season conditions. Actuator failure, control problems, and calibration error remain possible; without command, relay, reference, and capacity measurements, the log cannot identify the cause or actual cooling duty.

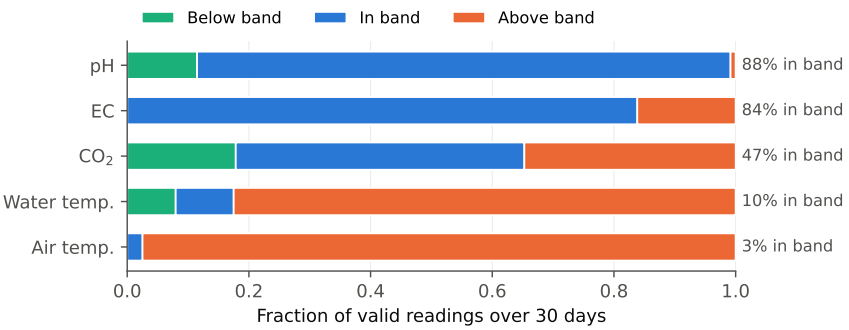


Figure 4: Fraction of valid readings below, within, and above each setpoint band over the deployment.

5.3. Fault-to-actuation audit

A naive controller replay issued 655 pH dosing commands, of which 125 (19.1%) used flagged readings, and 2149 CO_2 injection commands, of which 219 (10.2%) used flagged readings (Table 3, Fig. 5). The pH flags comprise 89 out-of-range values, 25 bound readings, and 11 heuristic spikes; the CO_2 flags comprise 74 out-of-range values and 145 spikes. A range check left 40 flagged pH commands: 28 bound readings and 12 spikes. Adding 32 s persistence removed all flagged pH and EC commands and reduced flagged CO_2 injections to 9 of 456 (2.0%), all nine attributed to heuristic spikes. Liveness and bound rejection produced the same executable CO_2 counts because high full-scale readings do not request injection in a raise-only loop.

These results correct an implementation error in the initial audit, which discarded unavailable directions only after they had consumed cooldown. Enforcing directions before authorization changes naive CO_2 counts from 1010/68 total/flagged to 2149/219 and range-plus-persistence counts from

441/7 to 456/9. The two-sided pH and EC results are unchanged. The separate frozen controlled benchmark deliberately remains bidirectional.

The legacy-gate proxy issued 43–72% as many commands as the range-and-persistence arm, depending on the channel, because its repeatability requirement corresponded to about 2–7 min of confirmation. It nevertheless authorized 13 CO_2 injections using out-of-range readings: after 50 consecutive rejections its reset accepts the current sensor value. These are proxy outputs, not evidence that the deployed firmware issued those commands.

Table 3: Executable replay commands and commands using flagged source readings (out-of-range, declared-bound, or heuristic-spike flags). Tot.: total commands; Flag.: commands using flagged readings. No actual command execution was logged.

	pH dosing		EC dosing		CO_2 injection	
Gating arm	Tot.	Flag.	Tot.	Flag.	Tot.	Flag.
Naive threshold	655	125	535	6	2149	219
Range check	593	40	535	6	2105	148
Range + persistence	417	**0**	526	**0**	456	9
Legacy-gate proxy	216	**0**	227	**0**	330	13
Range + persistence + liveness	417	**0**	526	**0**	456	9
Range + persistence + bound	417	**0**	526	**0**	456	9

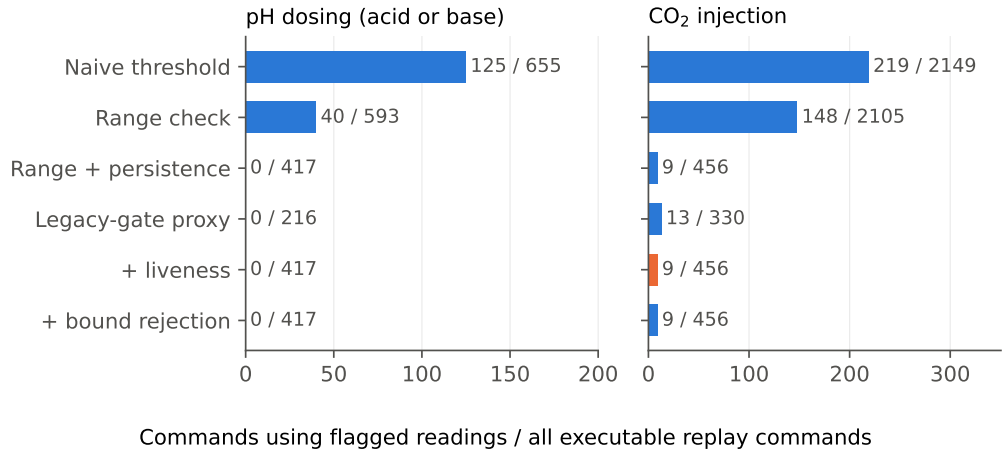


Figure 5: Executable replay commands using flagged readings, with counts relative to all executable commands under each arm. The flags do not establish physical harm.

The liveness test reduced replayed air-temperature cooling commands by 10% (3215 to 2891) and water-temperature commands by 9% (2698 to 2446). Coarsely quantized channels can repeat values during legitimate operation. These reductions are an authorization cost, not observed reductions in cooling or evidence of physical benefit.

5.4. Sensitivity of the audit finding

The complete grid falsified the stronger prediction that the range-and-persistence arm would remove every flagged pH authorization across timings (Table 4). There were zero to three flagged pH commands per setting, with nonzero counts in 54 of 108 pH attribution cells. With 32 s persistence and 64 s maximum age, shortening cooldown from 600 to 300 s admitted one bound-source command at the original eight-scale spike cutoff. EC remained at zero throughout. Relative to naive replay at the same timing and spike definition, flagged counts fell by 98.1–100% for pH and 87.1–99.7% for CO_2. These are whole-log descriptive reductions, not estimates of prevented harm. Rare air-temperature flags did not uniformly decrease; in one comparison a count rose from one to two as authorizations were rescheduled. The useful scope is a measured-log authorization tradeoff with explicit timing boundaries, rather than a guarantee that short persistence eliminates faulty decisions.

Table 4: Ranges across the complete post-hoc timing and spike-attribution grid for range plus persistence. Flagged counts use disjoint out-of-range, bound and spike categories. Each row summarizes 108 attribution cells from one deployment. Retention is relative to naive command counts at the same timing; it does not establish useful-action coverage.

Channel	Total commands	Flagged commands	Retention (%)
pH	207–829	0–3	50.7–74.7
EC	267–1032	0	97.4–99.1
CO_2	121–1576	1–68	9.8–53.2
Air temperature	1642–6275	0–2	98.5–99.8
Water temperature	1368–5288	0	98.7–99.9

5.5. Controlled benchmark and the locked LiveGate result

No method reached the 0.95 coverage target on validation. At the 0.5 and 0.8 targets, LiveGate did not meet the frozen criterion for best method (Table 5). At 0.5, LOF had the fewest undesirable authorizations (0.075

per trial versus 0.135 for LiveGate; paired difference 95% interval −0.148 to −0.004) but at much lower coverage (0.357). At 0.8, persistence only reached higher coverage (0.886) with fewer undesirable authorizations (0.117) than LiveGate (0.790 and 0.135). The liveness-only ablation had the lowest undesirable rate at 0.8 (0.104) but its evaluation coverage fell to 0.663. The AASVR-R measurement supervisor did not reach the 0.5 target. Figure 6 shows all validation operating points.

Table 5: Locked evaluation at the 180 s deadline. Settings were selected on validation; validation-infeasible methods are omitted. A dagger marks evaluation coverage below the corresponding target. Bold values are column extrema, not proof of an overall winner.

Target	Method	Coverage	Undesirable / trial
0.5	LOF	0.357†	**0.075**
	LiveGate, liveness only	0.560	0.081
	Kalman	0.594	0.098
	GLR	0.760	0.100
	Persistence only	0.795	0.110
	CUSUM	0.794	0.110
	Hampel / moving median	0.795	0.121
	EWMA	**0.801**	0.121
	LiveGate	0.505	0.135
	LiveGate, forecast only	0.704	0.163
	AASVR-R hold only	0.721	0.233
0.8	LiveGate, liveness only	0.663†	**0.104**
	Persistence only	0.886	0.117
	CUSUM	0.862	0.121
	LiveGate	0.790†	0.135
	Hampel / moving median	0.848	0.150
	EWMA	0.857	0.150
	LiveGate, forecast only	0.837	0.150
	Kalman	**0.910**	0.158
	GLR	0.854	0.165

The forecast-only ablation rejected 3.2% of uncorrupted readings at the 0.8 target, against none for persistence only, and had more undesirable authorizations (0.150 versus 0.117 per trial). Thus its extra rejection did not improve the aggregate authorization tradeoff in this benchmark. A fault and a legitimate change can produce the same local trace; additional context

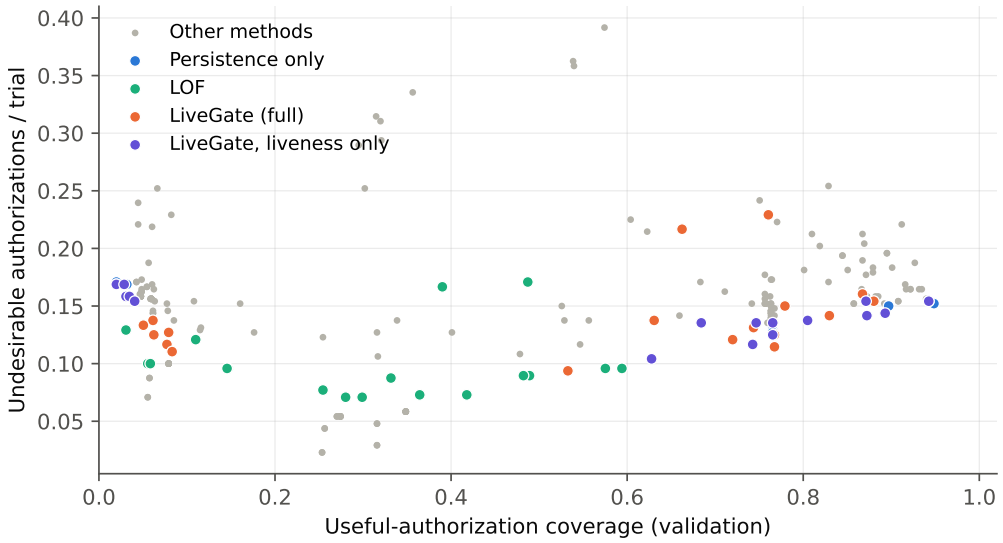


Figure 6: Validation operating points of all methods and settings. Points towards the lower right authorize more of the commands that are needed while issuing fewer that are not.

would have to be informative to distinguish them.

5.6. Stuck and saturated sensors

The original temporal-coincidence metric shows fewer authorizations during stuck and saturated intervals with liveness enabled (Fig. 7). At the 0.8 validation target, persistence only authorized during stuck samples 0.725 times per trial, compared with 0.225 for liveness only (69% fewer) and 0.500 for full LiveGate. During saturation, the corresponding rates were 0.975, 0.675 (31% fewer), and 0.650. Kalman reached 0.850; the other selected comparators matched persistence on this metric. These reductions do not by themselves show that fewer commands used corrupted source readings. They also accompany reduced overall coverage: 0.663 for liveness only versus 0.886 for persistence. The following analysis adds paired intervals, held-source attribution, and direct simple comparators.

5.7. Exploratory post-hoc variant

After the locked run, an exploratory variant floored the forecast scale at the sensor's reading resolution. All 66 prior evaluated rows reproduced exactly, and the variant's selected settings and results matched locked LiveGate

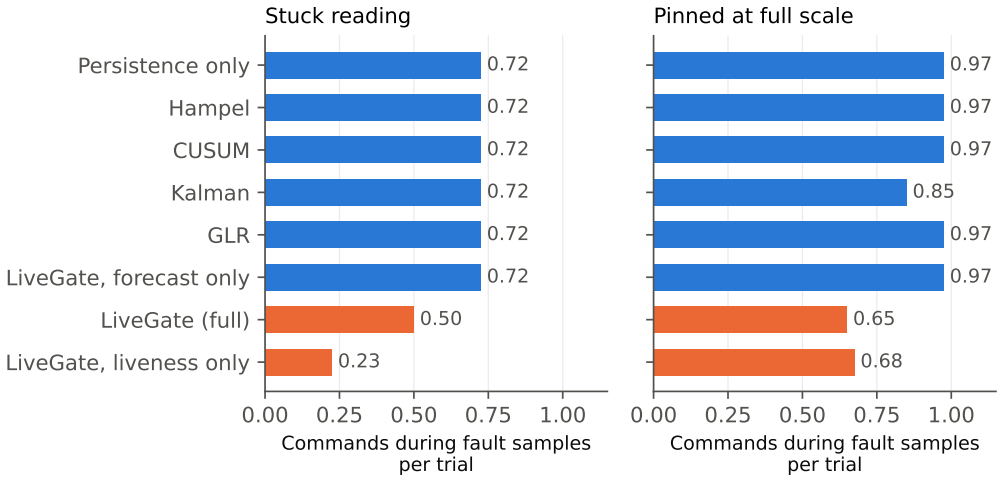


Figure 7: Authorizations occurring during injected-fault samples, per trial (locked evaluation, 0.8 validation target). This temporal metric does not identify the source of a held estimate. Liveness arms are highlighted.

at both feasible targets. The conformal threshold is calibrated on scores divided by the same scale, so rescaling moves the scores and threshold together. This variant did not address the forecast component's observed authorization tradeoff; it is retained as a disclosed null result.

5.8. Post-hoc direct-baseline comparison

A separately labelled follow-up reran the original methods and added two direct comparators: a fixed repeat cap and rejection at either physical bound. The repeat cap accepts the first R identical in-range readings, rejects subsequent repeats, and resets after a change, invalid reading, or logging gap. $R \in \{2, 4, 8, 16, 32, 64\}$ is shared across channels; the six-value grid is larger than LiveGate's three-value grid. Both comparators use the original timing grid and validation-only selection. All original evaluation summary rows and selections reproduced, while the frozen artifacts remained unchanged. The follow-up is post-hoc because the data and original results had already been examined.

Physical-bound rejection produced 0.000 commands per saturation trial using an injected-fault source, while its useful-command coverage in those trials was 0.073. Thus learned liveness was not unique in suppressing saturated source evidence. The fixed-cap gate and physical-bound gate improved both

Table 6: Post-hoc evaluation at the 0.8 validation coverage target. A dagger marks evaluation coverage below 0.8. Fault-source counts attribute each command to its last accepted reading, per trial. These are simulated authorizations.

Method	Coverage	Undesirable / trial	Fault-source commands Stuck	Saturation
Persistence only	0.886	0.117	0.725	0.975
LiveGate	0.790†	0.135	0.500	0.575
Learned liveness only	0.663†	0.104	0.150	0.675
Fixed repeat cap	0.849	0.117	0.675	0.975
Physical-bound rejection	0.798†	0.083	0.725	0.000

overall coverage and undesirable-command rate relative to full LiveGate at the selected 0.8 target. Physical-bound rejection also improved both relative to learned liveness-only, but its evaluation coverage (0.798) fell slightly below 0.8. Persistence-only had higher coverage than the fixed-cap gate at the same undesirable rate. These are point-estimate operating comparisons, not statistical superiority claims or evidence of field safety.

At target 0.8, validation selected a repeat cap of 64, 32 s persistence and 64 s maximum age for fixed repeat cap.

At target 0.8, validation selected 32 s persistence and 64 s maximum age for physical-bound rejection.

For stuck, the learned liveness-only arm's reduction relative to persistence was 69.0 [54.3, 85.7]% for commands during injected-fault samples, and 79.3 [71.0, 88.9]% for commands using injected-fault source readings (paired source-day 95% intervals in brackets).

Useful-command coverage in the same stuck trials fell from 0.982 with persistence to 0.273 with learned liveness. These reductions therefore do not establish better fault handling at matched useful-command coverage.

For saturation, the learned liveness-only arm's reduction relative to persistence was 30.8 [18.0, 46.2]% for commands during injected-fault samples, and 30.8 [18.0, 46.2]% for commands using injected-fault source readings (paired source-day 95% intervals in brackets).

Useful-command coverage in the same saturation trials fell from 0.918 with persistence to 0.409 with learned liveness. These reductions therefore do not establish better fault handling at matched useful-command coverage.

These two counters answer different questions: a command during a fault interval can still use a held clean reading. Neither measures physical ex-

ecution or damage. The estimates use eight evaluation windows on seven source-day clusters; settings remain fixed during resampling, and the secondary intervals have no multiplicity adjustment. The limited number of days makes generalization uncertain.

Table 7: Sensor-specific learned-liveness effects relative to persistence at target 0.8. Differences are method minus persistence. Fault-source differences are commands per trial; brackets give paired source-day 95% intervals. Coverage differences pool all scenarios.

Sensor	Coverage difference	Stuck difference [95% CI]	Saturation difference [95% CI]
EC	-0.167	-0.750 [-1.000, -0.400]	-0.750 [-1.000, -0.400]
pH	-0.231	-0.375 [-0.750, -0.091]	-0.375 [-0.750, -0.091]
Air temp.	-0.458	-0.750 [-1.000, -0.499]	0.000 [0.000, 0.000]
Water temp.	-0.067	-0.250 [-0.500, 0.000]	0.000 [0.000, 0.000]
CO_2	-0.071	-0.750 [-1.000, -0.429]	-0.375 [-0.714, 0.000]

The learned rule is itself a sensor-specific repeat threshold at fixed ε. Comparison with a global fixed cap does not establish superiority over all per-sensor thresholds. Benefits must be weighed against the useful-authorizations lost by rejecting legitimate repeated readings.

5.9. Crop outcome in the same season

In the same season, NFT plants produced a mean aerial fresh mass of 264.8 g (SD 35.0, $n = 7$), compared with 68.2 g (SD 19.5) in soil and 109.5 g (SD 23.3) in manual DWC (Table 8). These treatments differ in growing medium, enclosure, CO_2 enrichment, cooling, nutrient management, and automation at once, so the comparison describes the deployment context and does not isolate the effect of automation or of any gating rule.

Table 8: Harvest of the season analysed (mean ± SD, $n = 7$ plants per treatment). FMAP and DMAP: fresh and dry mass of the aerial part; TPL: total plant length; TNL: total number of leaves; NL10: leaves longer than 10 cm.

Trait	NFT (automated)	Soil	Manual DWC
FMAP (g)	264.81 ± 35.02	68.24 ± 19.46	109.49 ± 23.31
DMAP (g)	33.19 ± 8.24	8.63 ± 2.51	13.85 ± 2.70
TPL (cm)	72.52 ± 11.61	19.46 ± 3.54	20.12 ± 2.04
TNL	20.29 ± 2.87	9.86 ± 1.68	11.86 ± 1.95
NL10	13.00 ± 2.00	3.00 ± 1.15	7.57 ± 1.51

6. Discussion

6.1. Which faults matter for actuation

The audit separates conspicuous flags from flags that supply executable commands. Under naive replay, one in five pH authorizations used an out-of-range, bound, or spike-flagged reading. A range check with 32 s persistence removed these flagged-source authorizations in this log, while reducing total pH commands from 655 to 417. Without reference measurements, this does not establish that every suppressed command was unnecessary. High CO_2 full-scale readings requested no injection because that loop can only raise concentration. A pinned low reading, or saturation in a two-sided controller, can instead support repeated authorizations subject to cooldown. The same sensor signature therefore has different command consequences depending on actuator direction and supervisor state.

6.2. Why simple rules competed well

In this benchmark, a bias fault and a legitimate level shift can initially produce identical single-channel observations. This construction makes confirmation time a useful discriminator and helps explain why persistence performs well among the tested settings. It does not establish an optimal frontier for other deployments, faults, or informative multichannel models. Repeated readings provide another signature, but normal quantization can produce the same pattern. At fixed ε, the learned survival rule is itself a sensor-specific run-length threshold; its value must be judged against direct run-length and bound-rejection rules at comparable coverage, not only against detectors designed for changing signals.

6.3. Design guidelines

The results suggest six design checks. (i) Evaluate physical-range checks and short persistence with the actual actuator directions and cooldown; they removed all flagged pH and EC source commands in this log. (ii) Avoid resets that eventually accept any value after repeated rejections; the legacy-gate proxy admitted out-of-range CO_2 readings. (iii) Compare run-length and bound rejection against persistence at matched coverage before enabling liveness gating; coarse resolution can otherwise suppress useful cooling authorizations. (iv) Select sensor ranges with headroom and test humidity sensing near condensation; a pinned channel cannot resolve further changes. (v) Measure cooling capacity, actuator duty, and reference temperatures before diagnosing poor regulation or resizing equipment. (vi) Log commands, relay states, and independent references alongside readings. Those measurements are needed to connect authorization differences to physical execution and outcomes.

6.4. Limitations

The evidence comes from one system and one 30-day season. The audit counts authorizations under a specified common controller and an approximate legacy-gate proxy; original firmware and actual execution records are unavailable. Spike labels are heuristic, bound values are ambiguous, and missing intervals prevent full-season occupancy estimates. The synthetic benchmark uses fixed historical backgrounds, an idealized bidirectional controller, and only seven evaluation-day clusters. Its intervals describe resampling variability for those backgrounds and fixed settings, not prospective deployment uncertainty. Data and time blocks had already been analysed; the follow-up baselines were added after the frozen comparison and are explicitly post-hoc. The large CO_2 full-scale episodes lie in development and were excluded from liveness fitting by the declared-bound rule, but were not unseen in time. The crop comparison is confounded by simultaneous treatment differences. A prospective deployment with recorded commands, relay states, independent references, and a frozen configuration is needed to establish physical execution, avoided damage, or crop benefits.

7. Conclusions

The deployment audit links descriptive sensor flags to executable replay commands while respecting actuator direction and cooldown. Under naive

replay, flagged sources supplied 19.1% of pH commands and 10.2% of CO_2 commands; range checks with 32 s persistence reduced these to zero and 2.0% at the original timing. The sensitivity grid retained zero EC flags but admitted up to three flagged pH commands, so the zero-count pH result is setting-specific. The temperature traces establish poor observed regulation, with its cause unresolved. LiveGate did not outperform simple rules overall. The post-hoc comparison found that bound rejection removed saturated-source authorizations, while learned liveness reduced stuck-source authorizations at a substantial coverage cost. Neither result establishes a general advantage at matched availability. The engineering contribution is an explicit account of how reading validity, held-value age, actuator direction, and supervisor timing jointly affect authorization. Actual execution and agronomic benefits remain unmeasured.

Data and code availability

Code, configurations, the frozen and follow-up evaluation protocols, the measured 30-day sensor log, harvest data, and supporting results are publicly available at `https://github.com/MSIChowdhury/Design-of-an-IoT-enabled-Scalable-Closed-Hydroponics-System-with-Intelligent-Parameter-Control-and-P`. A labelled synthetic stand-in is provided for code-level smoke tests and is not used for reported findings.

Declaration of competing interest

The authors declare no competing interests.

Declaration of generative AI and AI-assisted technologies

AI-assisted tools were used to help write analysis code, draft text, and check consistency. The authors reviewed and are responsible for the content.

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